

RE07 Global Infrastructure Investment and Management

Week 1

Introduction to Infrastructure

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MPhil in Land Economy, Lent Term

What this week does

Week 1 has three jobs. It fixes what we mean by infrastructure, which is harder than it sounds and matters for everything that follows. It establishes the empirical case that infrastructure investment raises output, and the serious challenges to that case. And it introduces the claim that infrastructure is an asset class with distinctive investment characteristics, a claim the rest of the module tests rather than assumes.

A	Defining infrastructure	30 min
B	Infrastructure and economic development	40 min
C	The investment need and the gap narrative	30 min
D	Infrastructure as an asset class	45 min
E	Investors, vehicles and market structure	35 min
F	The iron law of megaproject management	40 min
G	Course mechanics and the group project	30 min

Where the module connects to the rest of the MPhil

Week	Real Estate Finance	Planning, Growth and Regeneration	Environmental Policy
1	RE01 s1, real estate as an asset class	PGR03 s2–6, location theory and agglomeration	EP02, market failure; EP06 s1–4, energy resources
2	RE01 s3, WACC as a regulator's allowed return	PGR01 s3–4, UK planning; PGR04, institutions and rent-seeking	EP02, asymmetric information; EP07 w4, adjudication
3	RE02 LT5–8, private vehicles, fees and risk	PGR05 s8a, financialisation; PGR07 s3, regeneration finance	EP06 s5–6, the valley of death; EP01 LT4, climate finance
4	RE01 s3–4, capital structure; RE03 w7, funding	PGR01 LT9, land value capture	EP02, instrument choice and liability rules
5	RE02 MT4, information and market efficiency	PGR07, the three industry seminars	EP07 w8, private governance
6	RE02 LT3–4, valuation methods and testing	PGR02 MT w2, policy evaluation; PGR05 s4, transport appraisal	EP02, valuation and evaluation; EP06, multicriteria analysis
7	RE02 LT1–2, net zero and climate risk	PGR05 s5, the green transition; PGR01 LT13, planning for net zero	EP06 throughout; EP01 LT7, climate litigation

RM01 lecture 5, on hedonic modelling and omitted variable bias, is common to every student and is used in weeks 1, 5 and 6. Full mapping in the module handbook.

What you already have

Where this week draws on modules you have taken, or are taking now. Nothing below is assumed: each idea is stated in full before the module is named.

Your programme	Where it was taught	What it gives this week
Real Estate Finance	RE01 session 1, real estate as an asset class. Sessions 8 and 9, portfolios and risk factor models. Session 5, indices and price dynamics. RE02 Michaelmas lecture 5, key metrics and property indices.	The asset class question asked about a different asset, and the machinery for answering it. Also the appraisal smoothing problem, which recurs in week 6.
Planning, Growth and Regeneration	PGR03 lectures 2 and 3, urban economics and location theory. Lecture 4, externalities. Lectures 5 and 6, agglomeration and spatial equilibrium. PGR02 Michaelmas week 1, the economic performance of British cities.	Transport cost is the primitive of location theory, and infrastructure is what sets it. This is the mechanism the aggregate growth regressions cannot see.
Environmental Policy	EP02, why there is environmental regulation, and market failure. EP06 lectures 1 to 4, global energy resources and technologies, energy access and energy poverty.	The market failure taxonomy that justifies public involvement, and the sector where the investment question is largest and most urgent.
Everyone	RM01 lecture 5, hedonic price modelling and omitted variable bias.	Why the correlation between infrastructure and growth is not the causal effect, which is the whole of block B.

Block A

Defining infrastructure

What the word has to carry, and why the boundary keeps moving.

Indicative length 30 minutes

The word has to do too much work

Infrastructure names a road, a fibre network, a hospital, a wind farm, a prison and a data centre. No single physical, economic or legal property is shared by all of them. What holds the category together is a family resemblance and, increasingly, a financial convention: infrastructure is what infrastructure funds buy.

That sounds glib. It is also close to how the market behaves, and it explains why the boundary of the asset class has moved so far in twenty years.

Economic and social infrastructure

Economic infrastructure

- Serves production directly and generally carries a user-facing revenue stream
- Transport: roads, rail, ports, airports
- Energy: generation, transmission, distribution
- Water and waste
- Digital: fibre, towers, data centres
- Usually charged for at the point of use, or through a regulated tariff

Social infrastructure

- Serves the delivery of public services and rarely carries a user charge
- Health: hospitals, primary care estates
- Education: schools, university estates
- Justice: courts, prisons
- Social housing and public administration estates
- Usually paid for by the state under an availability contract

Six properties commonly claimed for infrastructure

High sunk cost

Capital is committed up front and cannot be redeployed. A bridge has no alternative use and no resale market.

Long asset life

Thirty to a hundred years or more, which pushes the valuation weight into distant and poorly observed cash flows.

Natural monopoly

Minimum efficient scale often exceeds market size, so competition in the market is infeasible and regulation substitutes for it.

Inelastic demand

The service is close to essential, so volumes are less sensitive to price and to the economic cycle than for most goods.

Externalities

Costs and benefits fall on parties outside the transaction, which is the standard justification for public involvement.

Political salience

Prices and service standards are visible and contested, so the regulatory settlement is never permanently settled.

The properties are a checklist, not a definition

Presenting these six properties as the definition of infrastructure invites a mistake that runs through the practitioner literature. The properties are not jointly necessary and they are certainly not jointly sufficient. Assets are sold into infrastructure portfolios on the strength of two or three of them.

The useful question is not whether an asset is infrastructure. It is which of these properties it actually has, because each one maps onto a specific risk that has to be allocated to somebody in the contract.

The taxonomy investors actually use

Revenue model	Who bears demand risk	Typical assets	Return profile
Regulated	Shared, through the regulatory settlement	Water networks, electricity distribution, gas networks	Low volatility, returns set by the regulator on the asset base
Contracted	The offtaker, under a long-term contract	Availability PPPs, contracted renewables, rolling stock	Bond-like while the contract runs, then uncertain
Merchant	The asset owner	Toll roads without guarantees, merchant power, airports	Equity-like, correlated with the economic cycle

Greenfield and brownfield

Greenfield

- The asset does not yet exist
- Construction, completion and commissioning risk are live
- No operating history to underwrite against
- Higher expected return, wider distribution of outcomes
- Where the cost overrun evidence in Block F bites hardest

Brownfield

- The asset exists and is operating
- Construction risk is behind it; operating and regulatory risk remain
- A revenue record exists, though often a short one
- Lower expected return, narrower distribution
- Where most institutional capital actually goes

Where the boundary is being pushed

Data centres

Long-lived, capital-intensive, contracted to hyperscale tenants. No monopoly, rapid technological obsolescence, and a tenant concentration a utility would never accept.

Mobile towers

Sold out of telecom operators into infrastructure funds on the strength of contracted rents. The underlying technology cycle is far shorter than the asset life assumed.

Care homes and student housing

Social infrastructure in form, real estate in economics. Demand risk sits with the operator and the covenant is often weak.

Fibre to the home

Genuine network economics and high sunk costs, but competitive overbuild in several markets has destroyed the monopoly premise.

Renewable generation

Contracted output looks like infrastructure. Once the subsidy or power purchase agreement expires, the asset is a commodity price bet.

Rolling stock and aircraft

Mobile assets with a residual value market, which is closer to leasing than to infrastructure, yet routinely held in infrastructure vehicles.

Why the boundary problem is not academic

If the category expands to include whatever can be sold as infrastructure, then the empirical claims made about the category stop being reliable. Historical return series, correlation estimates and default statistics are all computed on a definition that has moved.

When an investor reports that infrastructure has delivered stable, inflation-linked returns with low correlation to equities, the first question is which assets were in the sample and whether they resemble what is being bought today. We return to this in Block D and again in Week 6.

The ownership and control spectrum

Arrangement	Who owns the asset	Who bears demand risk	Example
Public provision	The state	The state	A local authority road
Public corporation	The state, at arm's length	The corporation, backed by the state	A state-owned rail operator
Availability PPP	Special purpose vehicle, transferring on expiry	The state	A PFI hospital
Concession	Special purpose vehicle, transferring on expiry	The concessionaire	A tolled motorway concession
Regulated private	A private company, in perpetuity	Shared through the price control	An English water company
Full private	A private company	The owner	A merchant power plant

The sectors and what distinguishes them

Transport

Demand risk is severe and forecastable only within wide bounds. Network effects mean the value of a link depends on links you do not own.

Energy

Splits sharply between regulated networks, which are the archetype, and generation, which is increasingly a commodity business.

Water and waste

The strongest natural monopoly case, the heaviest regulation, and the most politically exposed pricing.

Digital

Fast technological change against long asset lives. The newest entrant to the category and the least tested.

Social

No user revenue, so the covenant of the paying authority is the whole credit. Political risk is the dominant risk.

What they share

Very little, physically. What they share is a contractual or regulatory structure that converts an asset into a revenue stream.

Block A discussion

Seminar questions

1. Which of the six claimed properties does a hyperscale data centre actually have, and does the answer change if the tenant is a single technology company?
2. A government wants to attract private capital into new water treatment. Should it structure the asset as regulated or contracted, and what does that choice cost it?
3. If infrastructure is defined by what infrastructure funds buy, what happens to the definition in a downturn?

Block B

Infrastructure and economic development

What the evidence supports, what it does not, and why the two are often confused.

Indicative length 40 minutes

The claim and the problem

Every infrastructure prospectus, ministerial speech and multilateral report opens with the same claim: infrastructure investment raises output. The claim is plausible, widely believed, and difficult to establish.

The difficulty is not measurement. It is identification. Rich countries have more infrastructure and also more of everything else. Governments build where growth is already expected. A correlation between infrastructure stock and output tells us almost nothing on its own.

Three ways the causal claim goes wrong

- Reverse causality. Output growth generates the tax revenue and the demand that produce infrastructure, so the arrow can run either way.
- Omitted variables. Institutional quality, human capital and openness all move with both infrastructure and output.
- Heterogeneity. The effect of a road in a congested corridor and a road in an empty one are not the same number, so a single pooled coefficient may describe neither.
- Each of these has a standard remedy, and none of the remedies is fully convincing on its own. The literature has converged on panel methods with instrument or dynamic-panel identification, and on being honest about what remains.

The output elasticity of infrastructure capital

0.07 to 0.10

Long-run elasticity of output with respect to a synthetic infrastructure index covering transport, power and telecommunications.

Calderón, Moral-Benito and Servén estimate a long-run aggregate production function on a large country panel, using a dynamic heterogeneous approach that allows the coefficient to differ across countries.

The estimate is statistically and economically significant and survives alternative dynamic specifications and alternative infrastructure measures. It is the most careful positive answer the literature has.

A ten per cent increase in the infrastructure stock buys roughly one per cent of additional output. That is real, and it is a good deal smaller than the rhetoric implies.

Aggregate elasticities hide the thing that matters

An elasticity is an average over projects that were built. It contains the successful bypass and the airport with no passengers, and it cannot separate them. It also says nothing about whether the marginal project, the one now under consideration, resembles the average.

For a country with a thin network, the marginal project may be far better than the average. For a country that has already built the obvious things, it will be worse. Selection, not volume, is where the returns are made or lost.

Negative

Risk-adjusted return on the typical infrastructure investment in the study sample.

Ansar, Flyvbjerg, Budzier and Lunn assemble the largest project-level dataset of its kind and reach a conclusion at odds with the growth literature. The typical investment in their sample fails to deliver a positive risk-adjusted return.

Their further claim is macroeconomic. Where investment is debt-financed and projects do not pay, the result is an accumulation of debt, monetary expansion and financial fragility rather than growth. China, on their evidence, delivers infrastructure no better than rich democracies do.

The paper is contested. It is also the most serious empirical challenge to the case for volume-led infrastructure programmes.

Reconciling the two findings

They measure different things

- Calderón measures the contribution of the infrastructure stock to aggregate output
- Ansar measures the ex post financial and economic performance of individual projects
- A stock can be productive while the marginal additions to it are not
- Aggregate data cannot see the projects that should not have been built

What follows for policy

- The case for infrastructure is a case for good infrastructure, which is not the same argument
- Appraisal and selection institutions carry more weight than headline investment targets
- A national target expressed in pounds or per cent of GDP is an input measure, and inputs are not outcomes
- This is the argument Week 2 develops through public investment management

Channels through which infrastructure affects output

- Direct productivity. Infrastructure enters the production function as an input, lowering the cost of producing everything else.
- Market access. Transport links expand the effective size of the market, which supports specialisation and scale.
- Agglomeration. Improved connectivity raises effective density, and density raises productivity through matching, sharing and learning.
- Human capital. Water, sanitation, electricity and digital access affect health and schooling outcomes, with long lags.
- Crowding in. Reliable networks reduce the private cost of investment, which changes private capital formation.
- The channels operate on very different time scales, which is one reason short-run evaluations of infrastructure programmes tend to disappoint.

The distributional question the growth literature does not answer

An elasticity is silent about who gains. A transport link that raises aggregate output can also concentrate activity in the larger of the two places it connects, leaving the smaller worse off than before. The same investment can raise national output and widen spatial inequality.

For a course taught in a land economy department this is not a footnote. It is the reason infrastructure appraisal in the United Kingdom has spent a decade arguing about whether benefit-cost ratios should be weighted by place.

Where the literature started

0.39

Output elasticity of public capital estimated by Aschauer on United States time series data.

Aschauer's finding launched the modern literature and was almost immediately doubted. An elasticity of 0.39 implies returns to public capital far above those to private capital, which is difficult to reconcile with any equilibrium account of investment.

The subsequent thirty years have been an extended argument about identification, and the estimates have fallen steadily as the identification has improved. Calderón and colleagues arrive at roughly a quarter of Aschauer's figure.

The trajectory is itself the lesson. Early estimates of the returns to a fashionable investment class are usually too high, and the correction comes from better identification rather than from new data.

Agglomeration is where land economists have the advantage

The largest single benefit in most United Kingdom transport appraisals is not time saved. It is the wider economic impact that follows from raising effective density: firms match with workers more easily, share suppliers and infrastructure, and learn from proximity.

This is standard urban economics, and it is also the most contested component of any business case, because the elasticity applied to effective density determines whether a scheme passes. A student who understands where that elasticity comes from can interrogate a business case more effectively than most people in the room.

Infrastructure and the development goals

- Six of the seventeen Sustainable Development Goals depend directly on infrastructure delivery, including clean water, affordable energy, and industry and innovation.
- The access framing differs from the growth framing. Access asks who is connected; growth asks what is produced. A programme can succeed on one and fail on the other.
- Access measures are also more tractable: connection rates are observable in a way that counterfactual output is not.
- The development finance institutions in Week 3 and Week 4 are mandated on access rather than on return, which explains behaviour that looks irrational from a commercial standpoint.

Block B discussion

Seminar questions

1. Calderón and colleagues estimate an output elasticity of about 0.08. Ansar and colleagues find the typical project destroys value. Can both be right, and what would have to be true for that?
2. If selection matters more than volume, what institutional arrangements would you want to see before approving a national investment programme?
3. Under what conditions does a new transport link make the smaller of the two places it connects worse off?

Block C

The investment need and the gap narrative

Where the trillion-dollar numbers come from, and what they can and cannot support.

Indicative length 30 minutes

Every report opens with a gap



The infrastructure investment gap is the most quoted number in the sector and the least interrogated. It appears in multilateral reports, fund marketing and government strategies, usually to two significant figures and usually without a method.

A gap estimate is not an observation. It is the difference between two constructed quantities: what would be spent on current trends, and what should be spent to meet a stated objective. Both halves are modelled, and the second depends entirely on the objective chosen.

How a gap estimate is built



- Choose an objective. Meeting projected demand, matching a peer group ratio of investment to GDP, or achieving a development goal each produce different answers.
- Model required investment. Usually a demand projection converted into physical capacity and then into cost, with unit costs taken from historical averages.
- Model trend investment. Extrapolate current spending, which is itself poorly measured in most countries.
- Take the difference and aggregate to 2030 or 2040. Small differences in assumed growth compound into trillions over a twenty-year horizon.
- Nothing in this procedure establishes that the investment would be worthwhile, only that it would be needed to hit the chosen target.

The most quoted figure

US\$15tn

Estimated shortfall between projected and required global infrastructure investment to 2040, on the Global Infrastructure Hub Outlook.

The Outlook models required investment at roughly ninety-four trillion dollars to 2040 across transport, energy, water and telecommunications, against projected investment on current trends of about seventy-nine trillion.

The modelled requirement is benchmarked on the performance of best-performing peer countries. Change the peer benchmark and the gap changes by trillions. Add the Sustainable Development Goals for water and electricity access and it grows again.

The figure is useful as an order of magnitude and misleading as a target.

What the gap narrative is used for

Legitimate uses

- Establishing an order of magnitude for policy discussion
- Comparing relative need across regions and sectors
- Identifying which sectors are furthest from a service standard
- Making the case that public balance sheets alone will not cover it

Uses that do not follow

- Inferring that private capital will earn a return by filling the gap
- Treating the gap as a pipeline of investable projects
- Reading a large gap as evidence of a large opportunity rather than of a large obstacle
- Using it to justify a specific project, which requires appraisal not aggregation

Why the gap persists where the need is greatest



If the need is largest in low and middle income countries, and capital is globally mobile, the gap should close. It does not. The obstacles are not a shortage of capital but the terms on which capital will travel: currency mismatch between local revenues and hard currency finance, political and regulatory risk, the absence of a bankable project pipeline, and the sovereign credit ceiling that caps what any project in the jurisdiction can be rated.

These are the subject of Week 3 and Week 4. Introduce them here so that the gap looks like a structural problem rather than a marketing opportunity.

Need is not evenly distributed

Sub-Saharan Africa

The largest gap relative to output, concentrated in electricity access, water and sanitation, and the road network. Also the thinnest domestic capital market.

Asia

The largest gap in absolute terms, driven by urbanisation and by power demand. Considerable domestic savings, unevenly intermediated.

Latin America

A long history of concession-based delivery and an equally long history of renegotiation, which is the Week 2 evidence base.

MENA

Capital is not the constraint. Institutional quality, regulatory predictability and the diversification agenda are.

Europe

Renewal rather than expansion, plus the investment implied by the energy transition. A maintenance problem misdescribed as a growth problem.

North America

An ageing stock, a well-documented maintenance backlog, and a municipal bond market that finances it in a way no other region replicates.

Public balance sheets and what constrains them

The standard argument for private participation is that governments cannot afford the investment. Stated that way the argument is weak, because a government that can borrow at the sovereign rate can almost always finance an asset more cheaply than a private consortium can.

The stronger arguments are about capability and incentive rather than money: whether the state can manage delivery, whether transferring construction risk to a party that can price it improves outcomes, and whether an off-balance-sheet structure is being used to evade a fiscal rule rather than to improve delivery. Engel, Fischer and Galetovic in Week 2 make this precise.

Block C discussion

Seminar questions

1. You are asked to produce an infrastructure gap estimate for a single country. What is the first decision you have to make, and how much does the answer depend on it?
2. A minister cites the global gap figure to justify a specific rail scheme. What is wrong with the inference?
3. If capital is mobile and the need is greatest in developing economies, why does the capital not go there?

Block D

Infrastructure as an asset class

The claimed investment characteristics, and what the evidence does to them.

Indicative length 45 minutes

What it takes to be an asset class

An asset class is not a marketing category. It is a group of investments whose returns are driven by a common set of economic factors, which behave similarly under stress, and which are not spanned by existing asset classes already in the portfolio.

The last condition is the demanding one. If infrastructure returns can be reproduced by a combination of equities, bonds and real estate, then infrastructure is a strategy rather than an asset class, and it should be priced and benchmarked accordingly.

Five claims made for infrastructure as an investment

Stable cash flows

Essential services with inelastic demand, so revenue is less cyclical than corporate earnings.

Inflation linkage

Tariffs indexed by regulation or contract, so real returns are protected when inflation rises.

Low correlation

Returns driven by regulatory and contractual factors rather than by equity market sentiment.

Long duration

Cash flows extending decades, matching the liabilities of pension funds and life insurers.

Illiquidity premium

Compensation for accepting an asset that cannot be sold quickly, earned by investors who do not need to sell.

The common structure

Every claim is a claim about the contract or the regulatory settlement, not about the concrete. That is where to test them.

Inflation linkage is contractual, not physical

A regulated water network is protected against inflation because a regulatory formula indexes the allowed revenue, not because water is inherently inflation-proof. A contracted solar farm is protected for the life of its power purchase agreement and not one day longer. A merchant toll road is protected only to the extent that the concession permits toll increases and traffic tolerates them.

The practical question for any asset is therefore narrow and answerable: which index, applied to what share of revenue, with what lag, and until when.

The tail risk that recent data conceals

1927

Start of the reconstructed US listed infrastructure return series used to test the low-risk claim.

Bianchi, Bornholt, Drew and Howard reconstruct US listed infrastructure index returns back to 1927 and analyse them in a Carhart four-factor framework. Low market beta and the value premium explain most of the return.

Their finding on risk is the one that matters for portfolio construction. Returns measured over recent decades understate long-run tail risk, which turns out to be comparable to that of a portfolio of US equities.

Some indices they examine are undesirable in a portfolio selection framework; others, notably broad and small utilities, do show diversification benefit. The asset class does not behave as one thing.

The mismatch between the story and the vehicle

10 years

Typical contractual life of the closed-end private funds through which most institutions hold infrastructure.

Andonov, Kräussl and Rauh document what institutional investors say they want from infrastructure, which is long-term stable cash flow, and then what they actually buy, which is exposure through finite-horizon closed private funds.

The cash flows delivered by those funds display volatility and cyclicalities similar to other private equity, and performance depends on quick deal exits in much the same way.

The asset may be a forty-year asset. The vehicle is a ten-year vehicle. Investors are exposed to the vehicle.

Listed and unlisted infrastructure

Listed

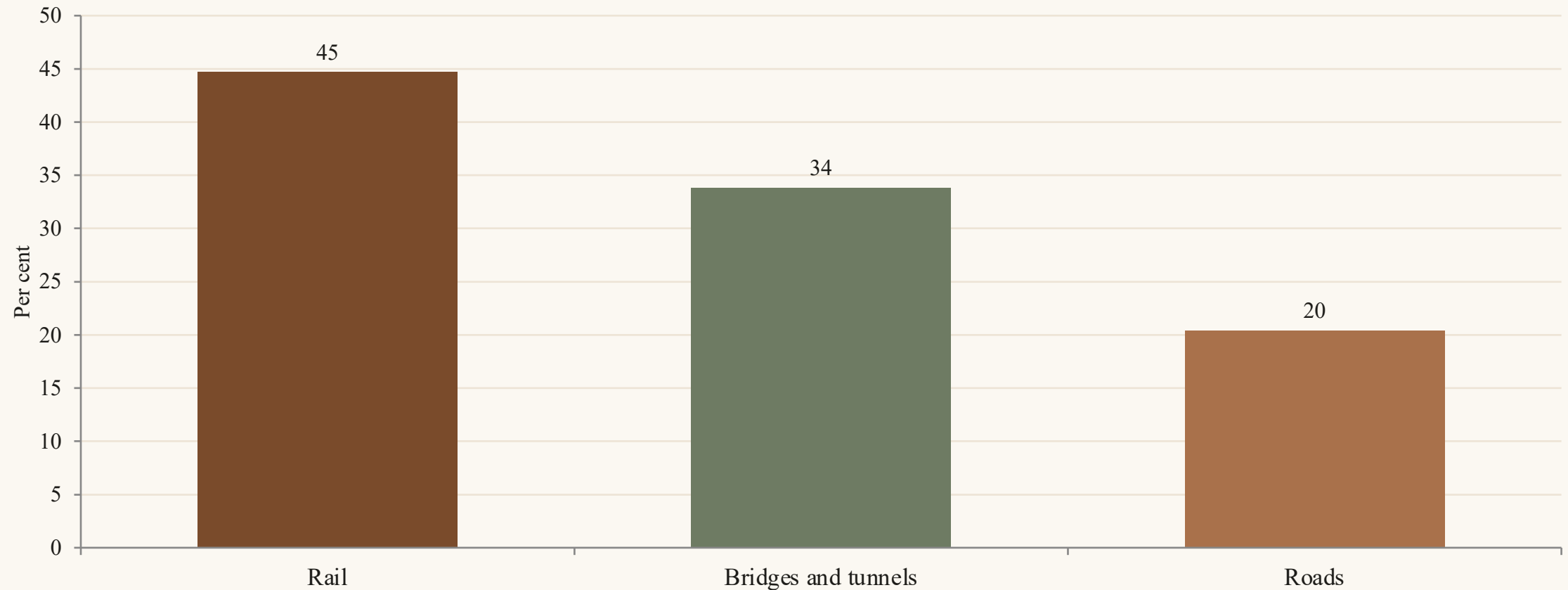
- Daily prices, genuine liquidity, low minimum investment
- Returns visibly correlated with equity markets, particularly in stress
- Company-level exposure includes management, leverage and unrelated businesses
- Reported volatility is real volatility

Unlisted

- Appraisal-based valuations, typically quarterly
- Reported volatility is smoothed by the appraisal process and understates the truth
- Direct asset-level exposure, with control rights
- Reported correlation with equities is partly an artefact of valuation method

Cost overrun by project type

Average cost overrun on 258 transport infrastructure projects, measured against the estimate made at the decision to build.



Flyvbjerg, B., Holm, M. S. and Buhl, S. (2002), Journal of the American Planning Association 68(3), 279–295. Sample of 258 transport projects worth about US\$90 billion.

The illiquidity premium is an assumption, not a measurement

Investors are routinely told that unlisted infrastructure earns a premium for illiquidity. The premium is difficult to observe because the return series against which it would be measured is itself constructed from appraisals rather than transactions.

What can be said is narrower. Investors who do not need liquidity give up little by forgoing it, so illiquid assets are cheap for them relative to investors who do. Whether the market prices that difference as a systematic premium is contested, and the evidence in unlisted infrastructure is thin.

Infrastructure debt and infrastructure equity

	Infrastructure debt	Infrastructure equity
Claim	Contractual, senior, secured on the project assets and its contracts	Residual, after operating costs, debt service and reserves
Return driver	Credit spread and the avoidance of default	Operating performance, refinancing and exit value
Evidence base	Long default and recovery series from the ratings agencies and consortium data	Short, appraisal-based, and dominated by a single vintage cycle
Who holds it	Banks, insurers, debt funds	Pension funds, sovereign wealth funds, listed and unlisted funds

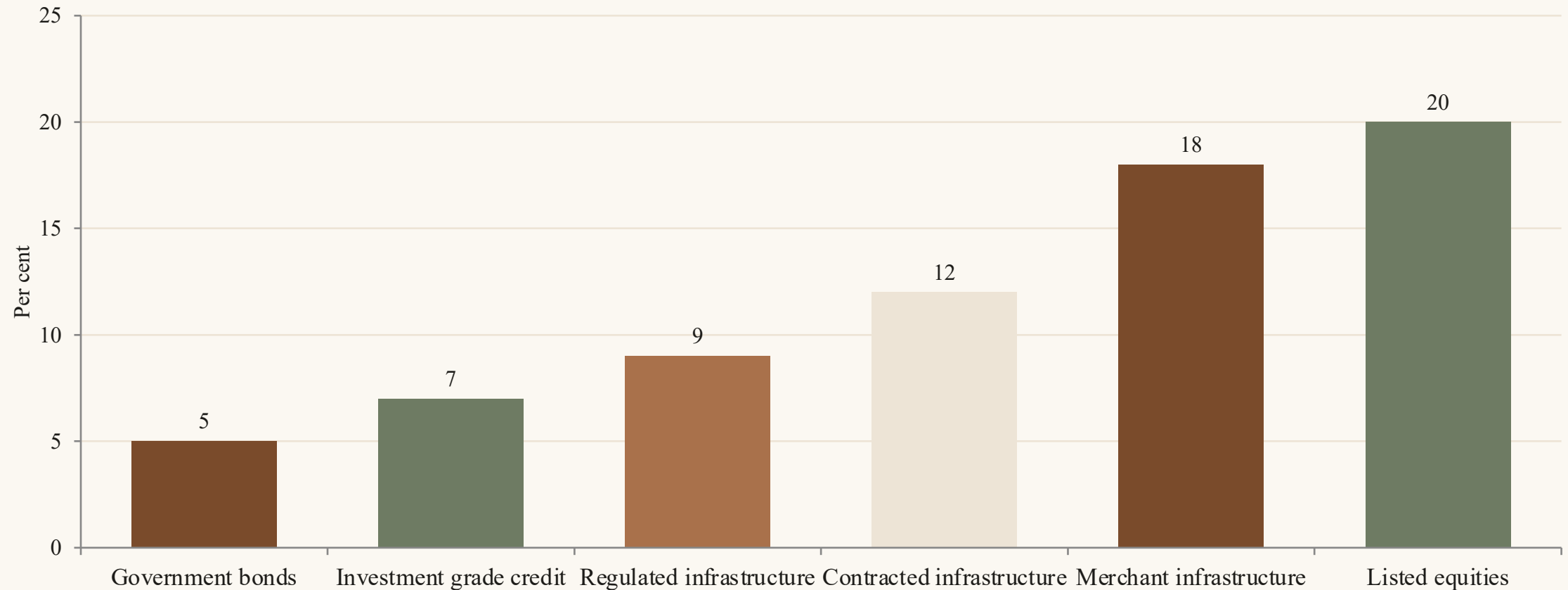
What survives the tests

The strong version of the asset class claim does not survive. Infrastructure returns are not uniformly stable, the recent record understates tail risk, and the vehicles through which the exposure is usually taken behave like private equity.

A weaker version does survive, and it is the version worth teaching. Certain infrastructure assets, held directly, with a regulated or contracted revenue base and a long holding period, do provide long-duration, partially indexed cash flow that is genuinely difficult to obtain elsewhere. The qualifications are doing real work in that sentence, and identifying which assets meet them is most of the job.

Where infrastructure is claimed to sit

The dispersion within infrastructure is wider than the distance between infrastructure and its neighbours.



Stylised illustration for teaching only. The figures are not estimates and are not drawn from a return series. The point of the exhibit is the ordering and its dependence on revenue model, not the magnitudes.

Correlation estimates deserve suspicion

The claim that infrastructure has low correlation with equities rests on return series that are, for unlisted assets, built from periodic appraisals. Appraisals are anchored on the previous appraisal and updated slowly, which smooths the series and mechanically reduces measured correlation and measured volatility.

The listed evidence, where prices are observed rather than estimated, shows correlation with equity markets that rises in stress, which is precisely when diversification is wanted. Whether the unlisted asset genuinely behaves differently or is merely measured differently is an open question, and it is the subject of Week 6.

The J-curve and why reported early performance misleads

A closed-end fund draws capital, pays fees, and invests before any asset has been improved or sold. Reported returns in the early years are therefore negative almost by construction, and reported returns in the later years benefit from the fees having already been paid.

For a course this matters in one specific way. Any comparison of infrastructure fund performance against a public market benchmark has to be made on the same basis, using a public market equivalent or an equivalent method, and comparisons made on internal rates of return alone are not meaningful.

Block D discussion

Seminar questions

1. A pension fund tells you it wants inflation-linked long-duration cash flow. Which of the five claims should it test first, and how?
2. If unlisted infrastructure valuations are appraisal-based, in what direction is reported correlation with equities biased, and why does that matter for asset allocation?
3. Andonov and colleagues find a mismatch between the asset and the vehicle. What would a vehicle designed for the asset actually look like, and why do so few exist?

Block E

Investors, vehicles and market structure

Who supplies the capital, through what structures, and why the structure shapes the outcome.

Indicative length 35 minutes

The main suppliers of capital

Pension funds

Long-dated, often inflation-linked liabilities. The natural holder in theory, constrained in practice by governance capacity and by size.

Insurers

Long liabilities and a regulatory capital regime that prices infrastructure debt favourably relative to equity.

Sovereign wealth funds

No liabilities in the ordinary sense, long horizons, and the scale to invest directly and take control positions.

Listed funds and corporates

Utilities, developers and conglomerates holding assets on balance sheet as strategic rather than portfolio investments.

Development finance institutions

Deploy capital where commercial investors will not, and increasingly to mobilise others rather than to lend alone.

Banks

Still the dominant provider of construction-phase debt, with an appetite shaped by capital regulation rather than by the asset.

Liability structure drives allocation

An investor allocates to infrastructure because of what it owes, not because of what infrastructure is. A mature defined benefit pension scheme paying inflation-linked pensions for forty years has a liability that regulated infrastructure matches unusually well. An insurer writing annuities has a similar problem and a different capital regime. An endowment with no fixed liabilities has no such need and buys infrastructure only if it expects a return premium.

Read any allocation decision back to the liability and the apparent inconsistencies in market behaviour resolve.

Routes into the asset

Route	Control	Fees	Minimum scale	Typical holder
Listed equity	None	Low	Very low	Any investor
Unlisted closed fund	None, subject to fund governance	Management fee and carried interest	Moderate	Mid-sized pension funds and insurers
Co-investment alongside a fund	Limited	Reduced or none	High	Large pension funds
Direct investment	Full or shared with consortium partners	Internal cost only	Very high	Sovereign wealth funds, large plans, corporates
Infrastructure debt	None, but with covenants and step-in rights	Low	Moderate	Insurers, debt funds, banks

The fund model and what it costs

The dominant vehicle is a closed-end limited partnership: a general partner raises capital from limited partners, invests it over an investment period, manages the assets, and returns capital through exits within a fixed life. The general partner charges a management fee on committed or invested capital and takes a share of profits above a hurdle.

The model is well suited to assets bought to be improved and sold. Its fit with assets bought to be held for forty years is poor, which is the structural point behind the Andonov finding in Block D. The fee load is the visible cost. The forced exit is the expensive one.

The direct investment model

- Large Canadian and Australian plans built internal teams and buy assets directly, alone or in consortium.
- The saving is not only the fee. Direct ownership removes the fund life, so the holding period can match the asset.
- The cost is organisational: an internal team capable of underwriting and governing a regulated utility is expensive and hard to retain.
- It is available only at scale. Below roughly the size at which an internal team can be justified, the fund route dominates.
- Consortium ownership creates its own governance problem, which is why shareholders agreements matter as much as concession agreements. Week 4 returns to this.

Market structure in one paragraph

Capital for infrastructure is abundant, concentrated in a small number of very large investors, and unevenly matched to the assets that need it. It congregates in brownfield assets in stable jurisdictions with established regulatory regimes, which is where the least of the investment need lies. The parts of the market where new capacity is needed most, and where construction and political risk are highest, are served by development institutions, export credit agencies and a shrinking group of specialist banks.

That asymmetry, rather than any absolute shortage of capital, is the central problem of infrastructure finance.

A worked example of fee drag

Take an asset expected to deliver eight per cent a year over a thirty-year life. Held directly, the investor receives eight per cent less internal cost. Held through a fund charging one and a half per cent on invested capital and twenty per cent of profits above an eight per cent hurdle, and sold after ten years into a market that may or may not be receptive, the investor receives materially less and also loses the twenty years of holding that made the asset attractive in the first place.

The arithmetic is straightforward and worth doing on the board. The conclusion is not that funds are indefensible, but that the fund route has to earn its cost through access to assets the investor could not otherwise reach.

What development finance institutions actually do

- Lend and invest where commercial capital will not, at pricing that reflects a development mandate rather than a required return.
- Provide guarantees and credit enhancement that move a project across a rating threshold that commercial investors are bound by.
- Absorb political risk that a private lender cannot diversify, which is the finding in Hainz and Kleimeier.
- Mobilise rather than merely lend: Broccolini and colleagues estimate that multilateral lending crowds in roughly seven dollars of bank credit per dollar invested over three years.
- Set environmental and social standards that then propagate through the syndicate, which is a form of regulation by lending.

Consolidation and its consequences

The manager side of the market has consolidated sharply. A small number of very large managers now raise the majority of capital committed to infrastructure, and several are divisions of firms whose primary business is private equity.

Two consequences follow. Fund sizes have grown faster than the supply of assets that meet the traditional definition, which is one force behind the boundary expansion in Block A. And the behavioural model of the buyer has shifted towards private equity practice, which is the mechanism behind the Andonov finding.

Block E discussion

Seminar questions

1. A mid-sized pension scheme cannot justify an internal infrastructure team. What is the best available structure for it, and what does that structure cost in return terms?
2. Why does capital concentrate in brownfield assets in stable jurisdictions when the stated need is elsewhere?
3. What would have to change for a ten-year fund to be an appropriate vehicle for a forty-year asset?

Block F

The iron law of megaproject management

What the delivery record shows, and why it belongs in the first week.

Indicative length 40 minutes

The iron law

Over budget, over time, over and over again.

Flyvbjerg, B. (2014), Project Management Journal 45(2), 6–19.

How often projects run over

Nine in ten

Share of megaprojects in Flyvbjerg's samples that exceed their approved budget.

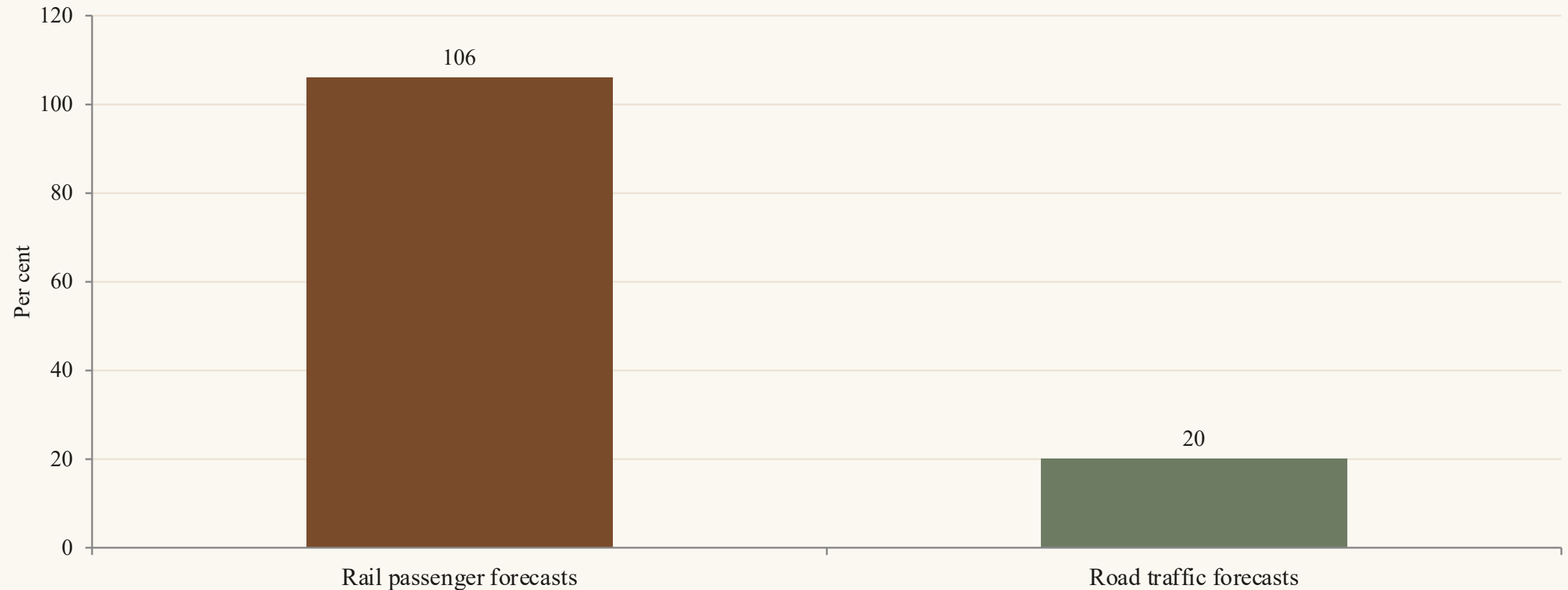
The claim is not that projects sometimes go wrong. It is that overrun is the normal case, that it has not improved over the several decades for which records exist, and that it holds across countries, sectors and procurement routes.

Flyvbjerg puts global megaproject spending at six to nine trillion dollars a year, or roughly eight per cent of world output. If nine in ten of those commitments are made on numbers that turn out to be wrong, the aggregate misallocation is large enough to matter macroeconomically.

That is the connection back to Block B, and to the Ansar argument about debt and fragility.

Demand forecasts are wrong in a predictable direction

Rail passenger forecasts are overestimated by an average of 106 per cent, and half of road projects miss forecast traffic by more than twenty per cent.



Flyvbjerg, B., Skamris Holm, M. K. and Buhl, S. L. (2005), Journal of the American Planning Association 71(2), 131–146. Sample of 210 projects in 14 nations. Rail figure is the average overestimate of passengers; road figure is the share of projects where actual traffic differs from forecast by more than twenty per cent in either direction.

Four explanations, and what each implies

- Technical error. Forecasting is hard and the models are imperfect. This would predict errors centred on zero and improving over time. Neither holds.
- Optimism bias. Planners are systematically over-optimistic, a psychological regularity documented well beyond infrastructure. This predicts bias but not its persistence once known.
- Strategic misrepresentation. Promoters understate costs and overstate demand because that is how projects get approved. This predicts persistent, directional bias that survives better methods, which is what the data show.
- Probabilistic causation. Overruns arise from the interaction of many small factors in a complex delivery system rather than from deception. This is the principal challenge to the Flyvbjerg account.
- Which explanation is right determines the remedy. Better models fix error. Reference class forecasting fixes bias. Only institutional checks fix misrepresentation.

245 dams

Sample used to establish the outside view for large hydropower projects.

Ansar and colleagues take an outside view: rather than modelling a project from its own components, compare it with the distribution of outcomes for the class of projects it belongs to.

On their sample, actual costs of large dams exceeded estimated costs by a large margin on average, and schedule overruns were routine. The method is now embedded in United Kingdom appraisal guidance in the form of optimism bias uplifts.

The technique transfers directly to investment underwriting. An investor presented with a base case can ask what the distribution of outcomes for this class of asset has been, and adjust accordingly.

Survival of the unfittest

If the projects that get approved are those with the most optimistic numbers, then the selection process rewards misrepresentation rather than merit. The project that should have been built loses to the project whose promoters were most willing to understate its cost.

The implication is uncomfortable and worth stating plainly. Systematically inaccurate appraisal does not merely produce disappointing projects. It produces the wrong projects, because it corrupts the ranking on which selection depends.

Why this belongs in Week 1

What it changes about the rest of the module

- A financial model is only as good as the cost and revenue forecasts underneath it, which we now know are biased
- Cover ratios calculated on a base case inherit the base case's optimism, which is the Week 6 problem
- Risk allocation in Week 4 is the contractual response to exactly these failures
- Appraisal and gateway institutions in Week 2 are the public response

What it should not become

- A counsel of despair. Some projects are delivered well and the block should name a few
- A claim that the private sector is immune. The same biases appear in privately promoted schemes
- A reason to avoid infrastructure. The evidence argues for better selection, not for less investment
- A single explanation. Love and colleagues offer a serious alternative and Week 6 gives it a hearing

The four sublimes

Political

The satisfaction of building monuments. Projects are visible, attributable and durable in a way that current spending is not.

Technological

The engineering appeal of the longest, tallest, fastest. The professionals who design projects want to design ambitious ones.

Economic

The business and employment created by construction, which accrues to identifiable firms and unions regardless of whether the asset is used.

Aesthetic

The pleasure of an iconic object. Bilbao and Sydney are cited in support of schemes with no comparable prospect.

What they have in common

Each supplies a constituency with an interest in the project proceeding, and none of them is an interest in the project being accurately costed.

Two cases worth keeping

The Channel Tunnel

- Delivered, and a genuine engineering achievement
- Construction costs ran far above the estimate on which investors subscribed
- Traffic fell well short of forecast for many years after opening
- Original equity was effectively destroyed while the asset itself continued to operate
- The distinction between a successful asset and a successful investment is the whole lesson

London 2012

- Delivered on time and within the revised budget, which is rare enough to be studied
- Systems integration was treated as a discipline in its own right rather than as project administration
- A separate delivery authority was created with a clear remit and a fixed deadline
- The immovable deadline was itself a governance device
- Flyvbjerg ch. 21 sets out the mechanism in detail

The distinction that matters for an investor

A project can be a triumph of engineering and a disaster for the people who financed it. The Channel Tunnel operates, carries millions of passengers, and did what its promoters said it would do physically. It also wiped out the investors who paid for it.

For this module the implication is precise. The question is never whether the asset will work. It is whether the cash flows will arrive in the amounts and at the times assumed, and who bears the loss if they do not. That question is the subject of Weeks 4, 5 and 6.

Block F discussion

Seminar questions

1. If cost underestimation is strategic rather than accidental, which institutional reforms would actually work, and which merely improve the forecasting method?
2. An investor is offered a greenfield project with a promoter's base case. What single adjustment would you make before underwriting it?
3. Does the iron law apply to privately promoted and privately financed projects to the same extent? What evidence would settle it?

Block G

Course mechanics and the group project

How the module runs, how it is assessed, and how teams are formed.

Indicative length 30 minutes

The eight weeks

Week	Topic	Practitioner
1	Introduction to infrastructure	Team formation and project launch
2	Institutional frameworks and stakeholder governance	Department for Transport
3	Private investment in infrastructure	Moody's and S&P Global Ratings
4	Infrastructure financing and deal structuring	International Finance Corporation
5	Intermediation and deal brokerage	Fund manager and principal investor
6	Performance and valuation	Academic and practitioner session
7	Sustainability, ESG and emerging trends	Central bank and climate risk
8	Presentations to the Global Infrastructure Advisory Panel	Full panel

Group project report, 50 per cent

- Teams analyse a live infrastructure investment issue supplied by a panel member
- Strategic and financial recommendations, submitted as a written report
- Presented to the Global Infrastructure Advisory Panel in Week 8
- Assessed on analysis, judgement under uncertainty, and the quality of the recommendation

Individual coursework, 50 per cent

- Two exercises, discussed in supervision
- The first builds and interrogates a project cash flow model
- The second applies a reference class adjustment to a base case and reports the effect on cover ratios
- Both are formative in sequence and summative in outcome

The Global Infrastructure Advisory Panel

The panel is a standing group of practitioners drawn from the Department's Global Land Economy network. Members deliver sessions, supply the live questions on which group projects are based, advise on case selection, and act as assessors for the Week 8 presentations.

The practical consequence for students is that the group project is not a simulation. The question comes from an organisation that wants an answer, and the recommendation is presented to people who will act on comparable questions in their own work.

Group project: what a good answer looks like

A recommendation

Not a survey of considerations. State what should be done, then defend it against the strongest objection.

A number

Every recommendation has a financial consequence. Quantify it, state the assumptions, and show what would change the answer.

A risk allocation

Name who bears each material risk under your proposal and why that party is the efficient bearer of it.

An honest range

Give the base case and the distribution around it. A single point estimate presented without a range will be marked down.

A stakeholder map

Identify who must agree, what each party needs, and where your proposal is politically vulnerable.

Brevity

A report that cannot state its recommendation in the first paragraph has not finished thinking.

Why the module brings three programmes together

The Department teaches three taught MPhils, and they currently meet in one place: Research Methods, which is a methods module rather than a subject. Infrastructure is the natural subject for that meeting, because a large project cannot be understood from any single position.

Someone who can price a project usually cannot get it consented. Someone who can get it consented usually cannot fund it. Someone who can assess its emissions and the policy instruments acting on it is rarely asked what a debt service cover ratio is. Every week of this module names the modules it draws on, grouped by programme, and the group project teams are composed across programmes on purpose.

What each programme brings, and what will be unfamiliar

Real Estate Finance

Brings pricing, capital structure, discounted cash flow, valuation under uncertainty, and the vocabulary of the capital markets.

Will find it unfamiliar that a project has to be consented before it can be financed, and that the regulatory settlement is renegotiated rather than fixed.

Planning, Growth and Regeneration

Brings the consenting process, who objects and on what grounds, how land value is captured, and how a public programme is evaluated after the fact.

Will find the capital stack and the cover ratio unfamiliar, and may be surprised that a lender's veto over a scheme is in practice stronger than a planner's.

Environmental Policy

Brings instrument choice, the economics of externalities, the energy system, and the law the transition actually runs on.

Will find it unfamiliar how a project is financed in detail, and what a commitment about future emissions is worth to a credit committee.

Team formation

- Teams of four to five, formed in this session, and composed across programmes. Aim for at least two of the three MPhils in every team, and all three where the numbers allow.
- A team drawn from a single programme will produce a report with a predictable blind spot, and it will be visible to the panel in Week 8.
- Mix quantitative capability deliberately as well. A team with nobody comfortable in a spreadsheet will struggle in Week 6.
- Each team nominates a coordinator, whose job is scheduling and submission rather than authorship.
- Project questions are allocated in Week 2 once the panel has confirmed the live issues for this year.
- Teams that cannot resolve an internal problem should raise it early. A dispute reported in Week 7 cannot be fixed.

Supervisions and how to use them

- Supervisions accompany the lectures and are where the individual coursework is discussed.
- Come with a specific difficulty rather than a general one. A supervision spent explaining what a cover ratio is has been wasted; one spent arguing about which ratio the lender should care about has not.
- The two formative exercises are set in Week 3 and Week 6 and discussed in the supervision that follows each.
- Reading is set per week in the module reading list. Two articles a week are required; the remainder support the group project.

Use of generative tools

The Department's policy applies in full and students should read it. For this module two practical points are worth stating in the first session. Any figure, citation or quotation that appears in submitted work is the student's responsibility to verify, whatever produced it, and fabricated citations are treated as an academic offence rather than as a technical error.

The second point is about learning rather than conduct. The judgement this module assesses, which is what to recommend under uncertainty and what to do when the numbers are contested, is precisely the part that cannot be delegated.

Reading for Week 2

- Delmon, ch. 5 on the institutional framework and ch. 6 on selecting transactions.
- Hart, O. (2003), Incomplete contracts and public ownership, *The Economic Journal* 113(486), C69 to C76. Seven pages, and the theoretical spine of the week.
- Engel, E., Fischer, R. and Galetovic, A. (2013), The basic public finance of public-private partnerships, *Journal of the European Economic Association* 11(1), 83 to 111.
- IMF, *Public Investment Management Assessment Handbook*, chapters 1 and 2, for the fifteen institutions of public investment management.
- Optional: Flyvbjerg (ed.) ch. 18 on the Norwegian quality assurance regime, which is the working example of what Block F argues is needed.

Block H

Reserve material

Slides held back from the main sequence, for use as the term develops.

Indicative length As required

Candidate cases for Week 1

Case	What it illustrates	Where it recurs
Channel Tunnel	A successful asset and a failed investment	Week 4 risk allocation
London 2012	Delivery under an immovable deadline and systems integration	Week 2 governance
Heathrow expansion	Consenting for nationally significant infrastructure	Week 2, guest session
Thames Tideway	A bespoke regulatory and financing structure for a single asset	Week 4 structuring
UK offshore transmission	Regulated revenue created by competitive tender	Week 3 investor appetite
A hyperscale data centre platform	The boundary of the asset class	Week 7 technology
A Latin American water concession	Renegotiation and regulatory capture	Week 2 Guasch evidence

An alternative opening, if the cohort is quantitative

Where the cohort arrives with a strong finance background, Block A can be compressed to the revenue model taxonomy alone and the time given to Block D. The definitional material then reappears in Week 4, where the contractual detail makes it concrete rather than abstract.

Where the cohort is drawn more from planning, law and policy, the reverse applies: keep Block A intact, compress Block D to the five claims and the two empirical tests, and rely on Week 6 to supply the technical content.

Sources drawn on for this deck

- Weber, B., Staub-Bisang, M. and Alfen, H. W. (2016), *Infrastructure as an Asset Class*, 2nd edn, Wiley, chapters 1 and 2.
- Abor, J. Y., Macomber, J., Arun, T. and Murinde, V. (eds) (2025), *The Routledge Handbook of Infrastructure Finance*, chapters 2, 3, 14, 15 and 16.
- Flyvbjerg, B. (ed.) (2017), *The Oxford Handbook of Megaproject Management*, chapters 1, 16, 17 and 21.
- Gatti, S. (2024), *Project Finance in Theory and Practice*, 4th edn, chapters 1 and 8.
- Delmon, J., *Private Sector Investment in Infrastructure*, chapters 1 and 2.
- Journal articles as cited on each slide. Full references in the module reading list.

Where Week 1 leaves us

Infrastructure is not one thing, and the properties that matter are contractual rather than physical.

The growth case is real but modest, and selection matters more than volume.

The asset class claim survives only in a qualified form, and the vehicle often does not match the asset.

The delivery record is poor in a way that is systematic, directional, and therefore correctable.